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1.3inch LCD HAT

1.3inch LCD HAT

1.3inch IPS LCD HAT, SPI interfaces

Introduction

Specification

- Operating voltage: 3.3V
- Interface: SPI
- LCD type: IPS
- Controller: ST7789VM
- Resolution: 240 (H) RGB x 240 (V)
- Display size: 23.4（H）x 23.4（V）mm
- Pixel size: 0.0975（H）x 0.0975（V）mm
- Dimension: 65 x 30.2(mm)

Pinout

PIN	Raspberry Pi Interface (BCM)	Description
KEY1	P21	KEY1GPIO
KEY2	P20	KEY2GPIO
KEY3	P16	KEY3GPIO
Joystick UP	P6	Upward direction of the Joystick
Joystick Down	P19	Downward direction of the Joystick
Joystick Left	P5	Left direction of the Joystick
Joystick Right	P26	Right direction of the Joystick
Joystick Press	P13	Press the Joystick
SCLK	P11/SCLK	SPI clock line
MOSI	P10/MOS	SPI data line
CS	P8/CE0	Chip selection
DC	P25	Data/Command control
RST	P27	Reset
BL	P24	Backlight

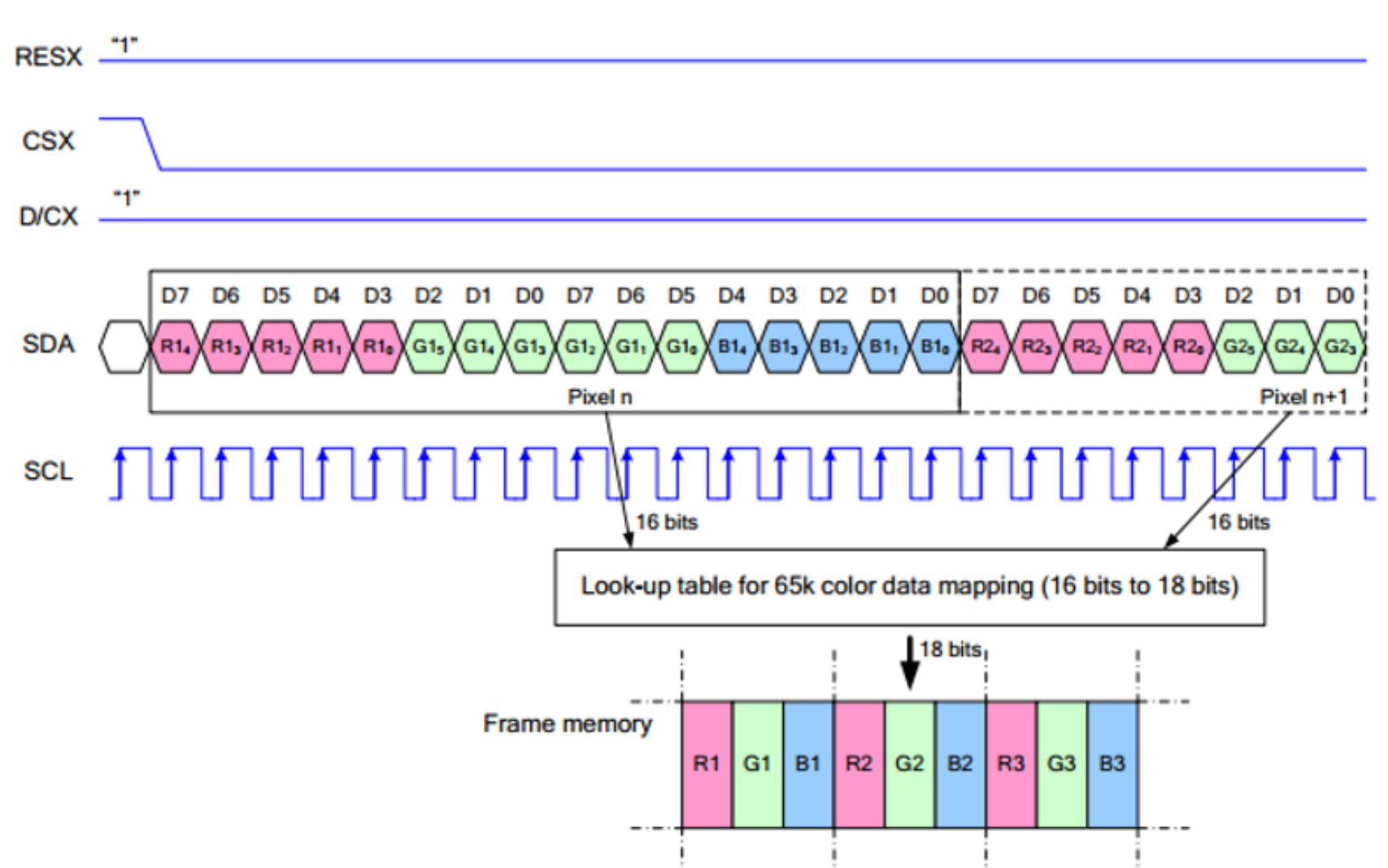
LCD and the controller

The ST7789VW is a single-chip controller/driver for 262K-color, graphic type TFT-LCD. It consists of 240 source line and 320 gate line driving circuits. The resolution of this LCD is 240 (H) RGB x 240(V), it supports horizontal mode and vertical mode, and it doesn't use all the RAM of the controller.

This LCD accepts 8-bits/9-bits/16-bits/18-bits parallel interface, that are RGB444, RGB565, RGB666. The color format used in demo codes is RGB565.

This LCD use a 4-line SPI interface for reducing GPIO and fast speed.LCD

Working Protocol


RESX
"1"
CSX
D/CX
"1"
SDA
D7 D6 D5 D4 D3 D2 D1 D0 D7 D6 D5 D4 D3 D2 D1 D0 D7 D6 D5 D4 D3 D2 D1 D0
Pixel n
Pixel n+1
SCL
16 bits
16 bits
Look-up table for 65k color data mapping (16 bits to 18 bits)
18 bits
Frame memory
R1 G1 B1 R2 G2 B2 R3 G3 B3

Note: Different from the traditional SPI protocol, the data line from the slave to the master is hidden since the device only has display requirement.

RESX Is the reset pin, it should be low when powering the module and be higher at other times; ;

CSX is slave chip select, when CS is low, the chip is enabled.

D/CX is data/command control pin, when DC = 0, write command, when DC = 1, write data

SDA is the data pin for transmitting RGB data, it works as the MOSI pin of SPI interface;

SCL worka s the SCLK pins of SPI interface.

SPI communication has data transfer timing, which is combined by CPHA and CPOL.

CPOL determines the level of the serial synchronous clock at idle state. When CPOL = 0, the level is Low. However, CPOL has little effect to the transmission.

CPHA determines whether data is collected at the first clock edge or at the second clock edge of serial synchronous clock; when CPHL = 0, data is collected at the first clock edge.

There are 4 SPI communication modes. SPI0 is commonly used, in which CPHL = 0, CPOL = 0.

Working with Raspberry pi

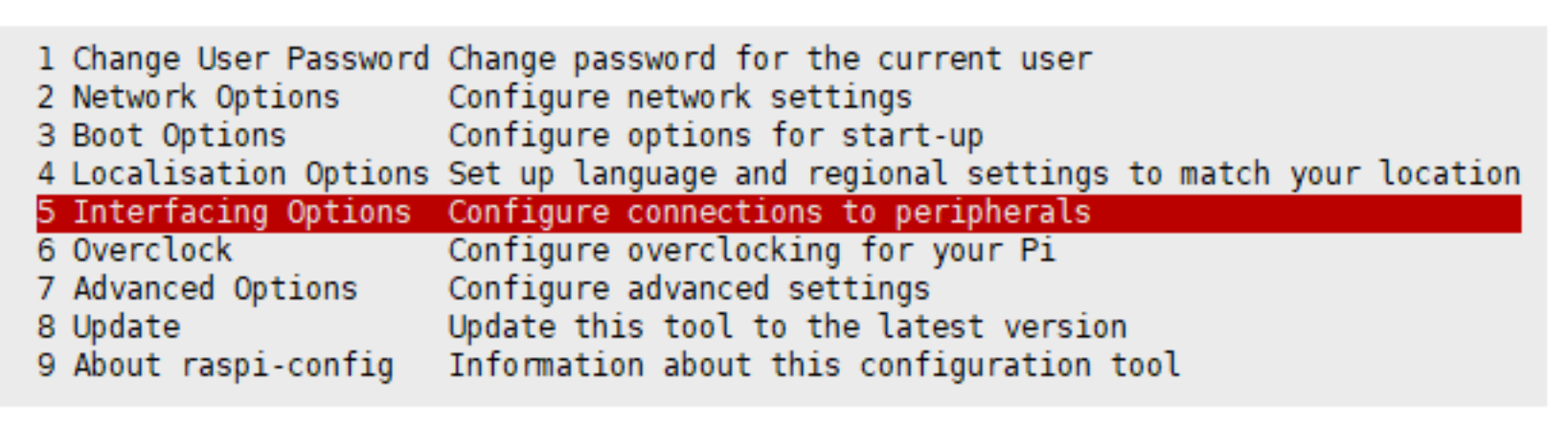
We provide C and python examples

Enable SPI interface

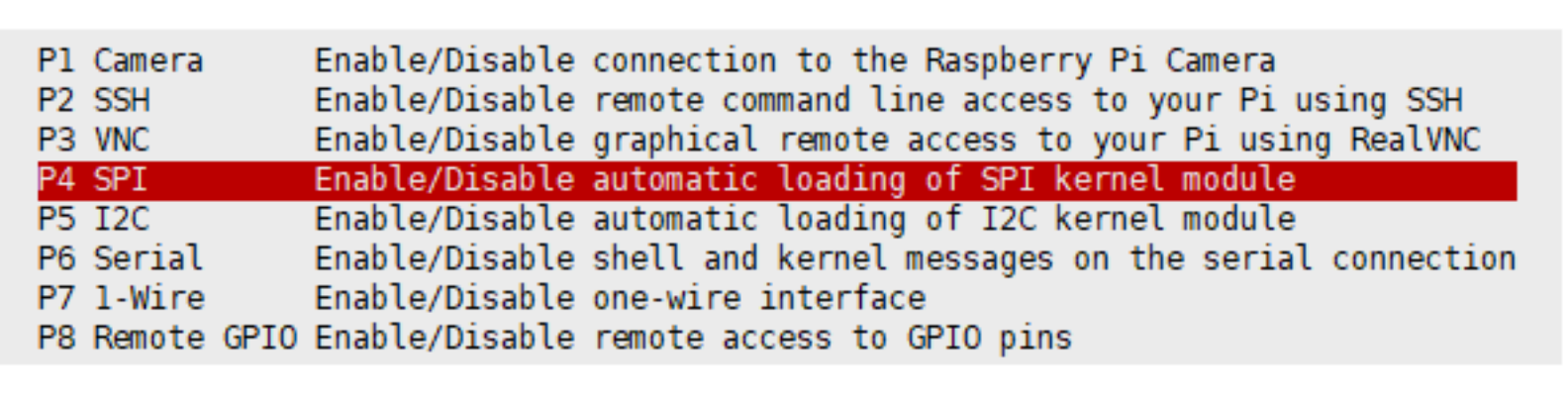
PS: If you are using the system of the Bullseye branch, you need to change "apt-get" to "apt", the system of the Bullseye branch only supports Python3.

- Open terminal, use command to enter the configuration page

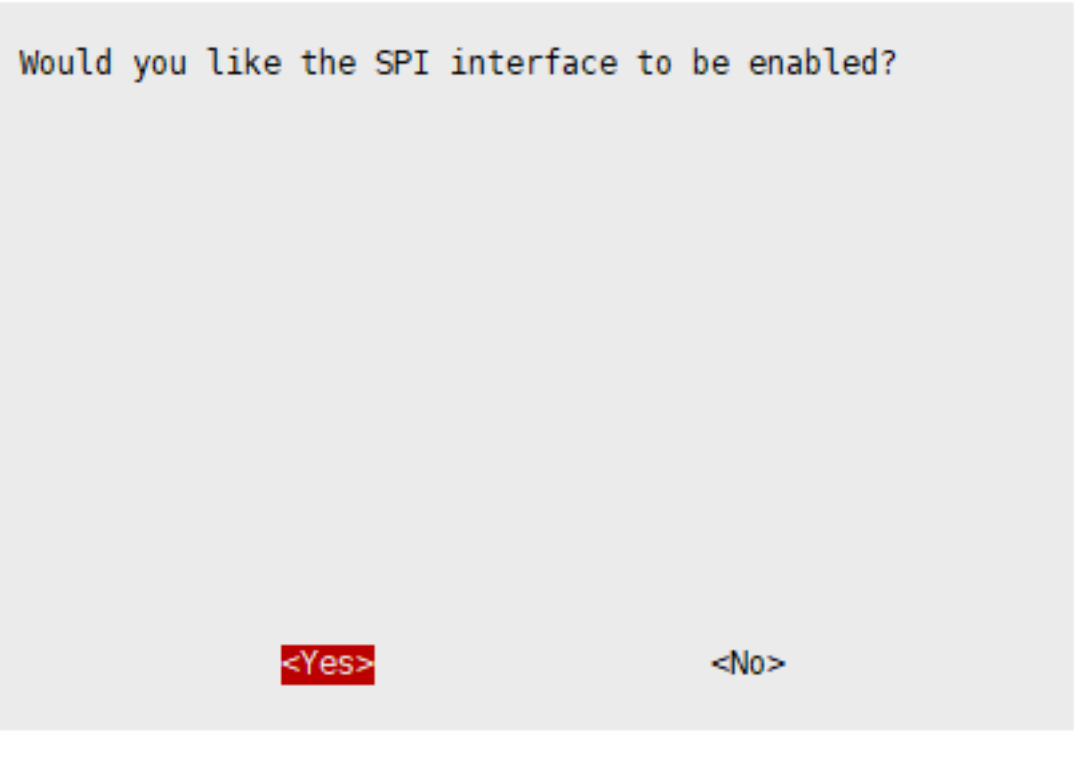
```
sudo raspi-config
Choose Interfacing Options -> SPI -> Yes  to enable SPI interface
```



```
1 Change User Password Change password for the current user
2 Network Options      Configure network settings
3 Boot Options         Configure options for start-up
4 Localisation Options Set up language and regional settings to match your location
5 Interfacing Options  Configure connections to peripherals
6 Overclock            Configure overclocking for your Pi
7 Advanced Options     Configure advanced settings
8 Update               Update this tool to the latest version
9 About raspi-config   Information about this configuration tool
```



```
P1 Camera      Enable/Disable connection to the Raspberry Pi Camera
P2 SSH         Enable/Disable remote command line access to your Pi using SSH
P3 VNC         Enable/Disable graphical remote access to your Pi using RealVNC
P4 SPI         Enable/Disable automatic loading of SPI kernel module
P5 I2C         Enable/Disable automatic loading of I2C kernel module
P6 Serial      Enable/Disable shell and kernel messages on the serial connection
P7 1-Wire      Enable/Disable one-wire interface
P8 Remote GPIO Enable/Disable remote access to GPIO pins
```



```
Would you like the SPI interface to be enabled?

<Yes>                <No>
```

Reboot Raspberry Pi：

```
sudo reboot
```

Please make sure that the SPI interface was not used by other devices, you can check in the /boot/config.txt

Install Library

If you use the bookworm system, you can only use lgpio library, bcm2835 and wiringPi can't be installed and used.

BCM2835

```
#Open the Raspberry Pi terminal and run the following command
wget http://www.airspayce.com/mikem/bcm2835/bcm2835-1.71.tar.gz
tar zxvf bcm2835-1.71.tar.gz
cd bcm2835-1.71/
sudo ./configure && sudo make && sudo make check && sudo make install
# For more, you can refer to the official website at: http://www.airspayce.com/mikem/bcm2835/
```

WiringPi

```
#Open the Raspberry Pi terminal and run the following command
cd
sudo apt-get install wiringpi
#For Raspberry Pi systems after May 2019 (earlier than that can be executed without), an upgrade may be required:
wget https://project-downloads.drogon.net/wiringpi-latest.deb
sudo dpkg -i wiringpi-latest.deb
gpio -v
# Run gpio -v and version 2.52 will appear, if it doesn't it means there was an installation error

# Bullseye branch system using the following command:
git clone https://github.com/WiringPi/WiringPi
cd WiringPi
. /build
gpio -v
# Run gpio -v and version 2.70 will appear, if it doesn't it means there was an installation error
```

lgpio

```
#Open the Raspberry Pi terminal and run the following command
wget https://github.com/joan2937/lg/archive/master.zip
unzip master.zip
cd lg-master
sudo make install

# You can refer to the official website for more: https://github.com/gpiozero/lg
```

Python

```
sudo apt-get update
sudo apt-get install python3-pip
sudo apt-get install python3-pil
sudo apt-get install python3-numpy
sudo pip3 install spidev
```

Download Examples

Open the Raspberry Pi terminal and run the following command:

```
sudo apt-get install p7zip-full -y
wget https://files.waveshare.com/upload/b/bd/1.3inch_LCD_HAT_code.7z
7z x 1.3inch_LCD_HAT_code.7z -r -o./1.3inch_LCD_HAT_code
sudo chmod 777 -R 1.3inch_LCD_HAT_code
cd 1.3inch_LCD_HAT_code
```

Run the demo

- C

```
cd c
make clean
make
sudo ./main
```

- For Raspberry Pi 4B and system version after raspbian_lite-2019-06-20, please set as following for normal input:

```
sudo nano /boot/config.txt
```

And then add the following line at the end of the config.txt.

```
gpio=6,19,5,26,13,21,20,16=pu
```

- python

```
cd python
sudo python main.py
sudo python key_demo.py
```

FBCP Driver

PS: FBCP is currently not compatible with 64-bit Raspberry Pi system, it is recommended to use 32-bit system.

The Framebuffer uses a memory area to store the display content, and changes the data in the memory to change the display content.

There is an open-source project on github: fbcp-ili9341. Compared with other fbcp projects, this project uses partial refresh and DMA to achieve a refresh rate of up to 60fps.

Compile and Run

```
cd ~
sudo apt-get install cmake -y
sudo apt-get install p7zip-full -y
wget https://files.waveshare.com/upload/f/f9/Waveshare_fbcp.7z
7z x Waveshare_fbcp.7z -o./waveshare_fbcp
cd waveshare_fbcp
mkdir build
cd build
```

If you are using 1.44inch_LCD_HAT:

```
cmake -DSPI_BUS_CLOCK_DIVISOR=20 -DWAVESHARE_1INCH44_LCD_HAT=ON -DBACKLIGHT_CONTROL=ON -DSTATISTICS=0 ..
```

If you are using 1.3inch_LCD_HAT:

```
cmake -DSPI_BUS_CLOCK_DIVISOR=20 -DWAVESHARE_1INCH3_LCD_HAT=ON -DBACKLIGHT_CONTROL=ON -DSTATISTICS=0 ..
```

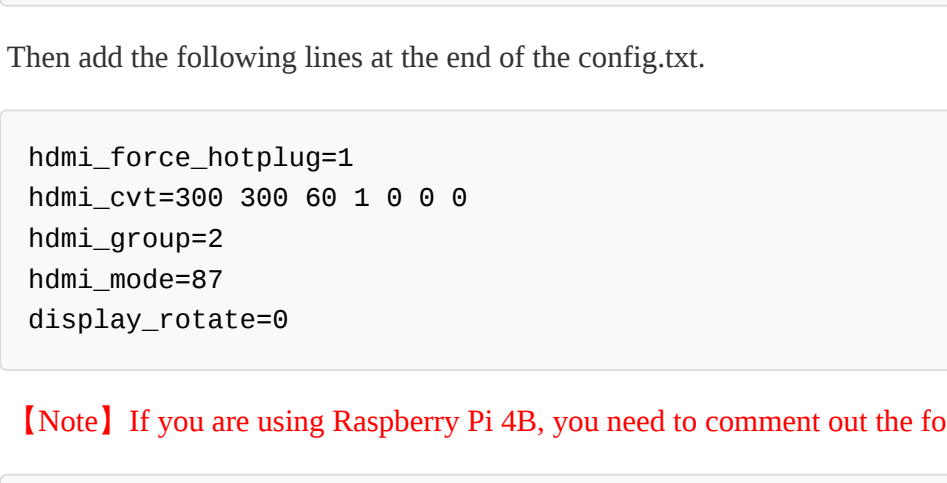
Then

```
make -j
sudo ./fbcp
```

Auto-start when Power on

```
sudo cp ~/waveshare_fbcp/build/fbcp /usr/local/bin/fbcp
sudo nano /etc/rc.local
```

And then add fbcp& before exit 0, as the picture below.



```
# Print the IP address
_IP=$(hostname -I) || true
if [ "$_IP" ]; then
  printf "My IP address is %s\n" "$_IP"
fi

fbcp&
exit 0
```

Set the Display Resolution

Set the user interface display size in the /boot/config.txt file.

```
sudo nano /boot/config.txt
```

Then add the following lines at the end of the config.txt.

```
hdmi_force_hotplug=1
hdmi_cvt=300 300 60 1 0 0 0
hdmi_group=2
hdmi_mode=87
display_rotate=0
```

【Note】If you are using Raspberry Pi 4B, you need to comment out the following lines on the [pi4] part. The modification is as below:

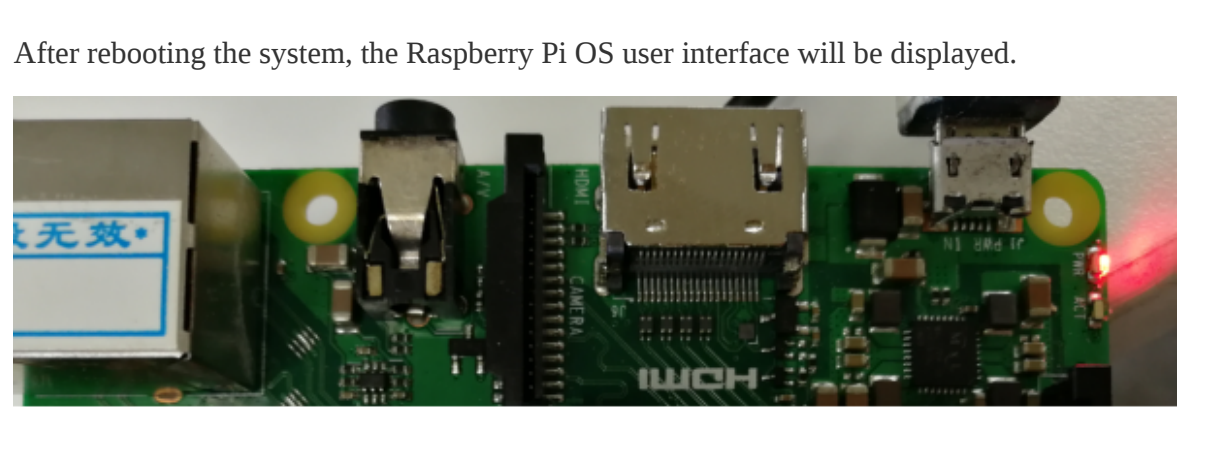
```
[pi4]
# Enable DRM VC4 V3D driver on top of the dispmanx display stack
#dtoverlay=vc4-fkms-v3d
#max_framebuffers=2
```

And then reboot the system:

```
sudo reboot
```

The final display effect is scaled and displayed on the 1.3-inch LCD in proportion. The setting of the resolution here should be slightly larger than the LCD resolution, too high resolution will cause the font display to be blurred.

After rebooting the system, the Raspberry Pi OS user interface will be displayed.



Analog Mouse

There are a joystick and three buttons on the panel of the module, which we can use to control the mouse of the Raspberry Pi

- Install the library, then download and run the demo

```
#python2
sudo apt-get install python-xlib
sudo pip install PyMouse
wget https://files.waveshare.com/upload/d/d3/Mouse.7z
7z x Mouse.7z
sudo python mouse.py
#python3
sudo apt-get install python3-xlib
sudo pip3 install PyMouse
sudo pip3 install unix
sudo pip3 install PyUserInput
wget http://www.waveshare.net/w/upload/d/d3/Mouse.7z
7z x Mouse.7z
sudo python3 mouse.py
```

- 【Note】The mouse.py needs to run under the graphical interface, which will not run under the SSH login. You can skip this step directly, the Pi will run the demo automatically by booting up.
- If you are using the Raspberry PI 4B and the version of your image is after raspbian_lite-2019-06-20, you need to set as below:

```
sudo nano /boot/config.txt
```

And then add the following line at the end of the config.txt.

```
gpio=6,19,5,26,13,21,20,16=pu
```

Use the joystick to move up, down, left, and right, you can see that the mouse is moving.

- Set the auto-start when power on

```
cd .config/
mkdir autostart
cd autostart/
sudo nano local.desktop
```

And then add the following lines at the end of the local.desktop

```
#python2
[Desktop Entry]
Type=Application
Exec=python /home/pi/mouse.py
#python3
[Desktop Entry]
Type=Application
Exec=python3 /home/pi/mouse.py
```

Reboot the system, you can use the buttons to control the mouse.

```
sudo reboot
```

Others

For connecting to the game machine, you can refer to the following links:

https://www.sudomod.com/forum/viewtopic.php?f=11&t=5371&start=10

https://pi0cket.com/guides/tiny-software-for-tinypi/#more-99

Resource

Document

- User Manual
- Schematic
- Display Desktop of Pi Study Manual

Demo

- Demo code
- Analog mouse

Relate Resources

- Pisugar case for 1.3inch LCD HAT

Software

- lcd
- Image2Lcd

Datashee

- ST7789VM

FAQ

Question: Setting fbtft does not recognize fb1, or the screen does not display?

Answer:

```
options fbtft_device name=flexfb gpios=reset:27,dc:25,cs:8,led:
24 speed=40000000 bgr=1 fps=60 custom=1 height=240 width=240
options flexfb setaddrwin=0 width=240 height=240 init=-1,0x11,-2,120,-1,0x36,0x70,-
1,0x3A,0x05,-1,0xB2,0x0C,0x0C,0x00,0x33,0x33,-1,0xB7,0x35,-1,0xBB,0x1A,-1,0xC0,0x2C,-
1,0xC2,0x01,-1,0xC3,0x0B,-1,0xC4,0x20,-1,0xC6,0x0F,-1,0xD0,0xA4,0xA1,-1,0x21,-
1,0xE0,0x00,0x19,0x1E,0x0A,0x09,0x15,0x3D,0x44,0x51,0x12,0x03,0x00,0x3F,0x3F,-
1,0xE1,0x00,0x18,0x1E,0x0A,0x09,0x25,0x3F,0x43,0x52,0x33,0x03,0x00,0x3F,0x3F,-1,0x29,-3
```

Note: these two statements start with OPTIONS.

Question: Why is the program-controlled after setting fbtft, and the error is reported that there is no SPI device

Answer:

These are two control methods. After setting to fbtft control, the SPI is occupied by the system, and the screen can no longer be controlled by operating the IO. You must block the /etc/module and change.

```
flexfb
fbtft_device
```

to

```
#flexfb
#fbtft_device
```

Then restart the raspberry pi

Question: How to play a video?

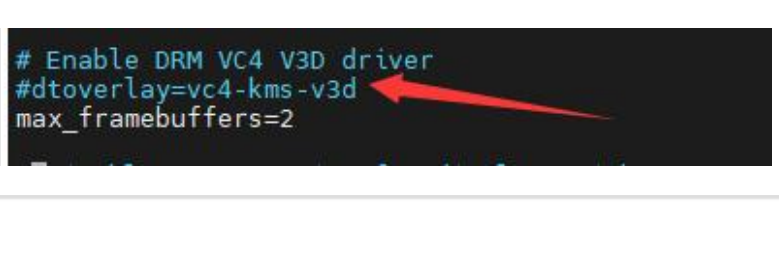
Answer:

To play a video, you must first set fbtft, and set fbtft to display the desktop through f, and then run the play video command to display.

Question: When executing fbcp file, vc_dispmanx_display_open failed!?

Answer:

Just comment out the kms driver.



```
# Enable DRM VC4 V3D driver
#dtoverlay=vc4-kms-v3d
max_framebuffers=2
```

Support

Technical Support

If you need technical support or have any feedback/review, please click the **Submit Now** button to submit a ticket, Our support team will check and reply to you within 1 to 2 working days. Please be patient as we make every effort to help you to resolve the issue.

Working Time: 9 AM - 6 AM GMT+8 (Monday to Friday)

Submit Now